

**Octopus Throughput Update:
SB3 SPEEDUP status + greedy baseline**

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Environment throughput improved by $2.73\times$ after SPEEDUP_PLAN.

Run	Steady fps	Overall fps
SB3 pre-SPEEDUP	1,009	1,050
SB3 post (C1+C2+C3)	2,757	2,791

Cache behavior

First run cache prep: 379.5s
Subsequent runs: $\sim 3s$

- ▶ Chunks 1–3: flat arrays, disk-backed precompute cache, Numba kernels.
- ▶ Throughput gain vs. pre-SPEEDUP: **$2.73\times$** .
- ▶ Post-speedup steady windows stay in a tight 2,533–2,959 fps band.
- ▶ Same benchmark setup: 300s env-only run, 32 envs, reward variant A.

Interpretation

Environment stepping is no longer the dominant bottleneck in training.

Training snapshot: env is fast, end-to-end still gated elsewhere.

Run	Steps	Wall time	End-to-end fps
JAX (scripts/train_jax.py)	50,016	72.8s	687
SB3 (scripts/train_rl.py)	50,176	58.0s	865

- ▶ Matched high-level settings: reward_variant=current, lookahead_window=200, reward_lambda=0.2, n_envs=32.
- ▶ Relative to old run7 training baseline (~ 53 fps), post-speedup env-only throughput is $\sim 52.8\times$ higher.
- ▶ Current 50k snapshot shows SB3 is $\sim 1.26\times$ faster than JAX end-to-end.

Greedy held-out baseline remains weak on both traces.

Trace (short)	Avg	Min	Max	Std
LVL01-troundgrt5m	-18.4948	-19.1440	-18.1463	0.211
AMS20-tround	-6.6626	-7.0170	-4.5441	0.609

- Metrics are over 50 greedy iterations per held-out trace.
- Both traces use the same 96-host / 192-MPD topology.

Takeaway

Greedy provides a stable but low-quality floor for held-out evaluation.

Next check

Compare these rewards directly against latest RL checkpoints on the same traces.

Held-out comparison: RL run7 vs latest greedy run.

Trace	Policy run	Pooling ratio mean	Savings mean
LVL01-troundgrt5m	RL (run7)	1.0601	-0.0601
LVL01-troundgrt5m	Greedy (latest)	1.0330	-0.0330
AMS20-tround	RL (run7)	0.9823	0.0177
AMS20-tround	Greedy (latest)	0.7215	0.2785

- ▶ RL and greedy were evaluated with reward variant A (reward_variant=current, $\lambda = 0.2$).
- ▶ New greedy run is better than run7 RL on both held-out traces in this latest measurement.
- ▶ Gap is small on LVL01 and large on AMS20 (~ 0.26 pooling-ratio improvement for greedy).
- ▶ Next step: Run 6 ablations and compare against greedy.